

Exercise Tolerance in Intermittent Cycling: Application of the Critical Power Concept

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ABSTRACT

CHIDNOK, W., F. J. DIMENNA, S. J. BAILEY, A. VANHATALO, R. H. MORTON, D. P. WILKERSON, and A. M. JONES. Exercise Tolerance in Intermittent Cycling: Application of the Critical Power Concept. *Med. Sci. Sports Exerc.*, Vol. 44, No. 5, pp. 966–976, 2012. **Purpose:** This study tested the relevance of the critical power (CP) model for explaining exercise tolerance during intermittent high-intensity exercise with different recovery intensities. **Methods:** After estimation of CP and W' from a 3-min all-out test, seven male subjects completed, in randomized order, a cycle test to exhaustion at a severe-intensity constant-work-rate (S-CWR) and four cycle tests to exhaustion using different intermittent (“work–recovery”) protocols (i.e., severe–severe (S–S), severe–heavy (S–H), severe–moderate (S–M), and severe–light (S–L)). **Results:** The tolerable duration of exercise in S-CWR was 384 ± 48 s, and this was increased by 47%, 100%, and 219% for S–H, S–M, and S–L, respectively (all $P < 0.05$). Consistent with this, compared with S-CWR (22.9 ± 7.4 kJ), the work done above the CP was significantly greater by 46%, 98%, and 220% for S–H, S–M, and S–L, respectively (all $P < 0.05$). The slope of the relationship between $\dot{V}O_2$ and time was significantly reduced for S–H, S–M, and S–L (0.09 ± 0.02 , 0.09 ± 0.01 , and 0.07 ± 0.02 L·min^{−2}, respectively) compared with S-CWR (0.16 ± 0.03 L·min^{−2}, $P < 0.05$). In addition, the slope of the relationship between integrated EMG and time showed a systematic decline for S–H, S–M, and S–L compared with S-CWR ($P < 0.05$). **Conclusions:** These results indicate that, when recovery intervals during intermittent exercise are performed below the CP, exercise tolerance is improved in proportion to the reconstitution of the finite W' . The enhanced exercise tolerance with the lower-intensity recovery intervals was associated with a blunted increase in both $\dot{V}O_2$ and integrated EMG with time. **Key Words:** CRITICAL POWER, W' , $\dot{V}O_2$ KINETICS, INTERVAL TRAINING, FATIGUE

The physiological responses to constant-work-rate (CWR) exercise are dictated by the intensity domain in which an individual is exercising (16,30,41,42). During moderate-intensity exercise (i.e., exercise performed below the lactate or gas exchange threshold, GET), a steady-state $\dot{V}O_2$ is typically achieved within 2–3 min. During exercise above the GET, however, development of the so-called $\dot{V}O_2$ “slow component” results in the attainment of a delayed and elevated steady state (heavy-intensity exercise) (40,41) or a continued rise in $\dot{V}O_2$ until $\dot{V}O_{2peak}$ is achieved, with the latter portending the limit of tolerance (T_{lim}), that is, severe-intensity exercise (14,30,42). Blood [lactate] (where square brackets denote concentration) responses also discriminate exercise intensity domains: for moderate exercise, blood

[lactate] remains close to resting values; for heavy exercise, an elevated but stable blood [lactate] is maintained; and for severe exercise, blood [lactate] rises inexorably until the limit of tolerance is reached (30,42). Finally, there is evidence that Type II fiber recruitment (23) and neuromuscular activity, as reflected by the integrated EMG (iEMG [8,38]), increase with time during severe exercise in concert with the development of the $\dot{V}O_2$ slow component (15).

The asymptote of the hyperbolic relationship between power output (P) and the limit of tolerance during high-intensity exercise (e.g., the so-called “critical power” during cycling; CP) marks the boundary between the heavy and severe domains and, therefore, represents an important physiological threshold that approximates the so-called maximal lactate steady state (31,32). The CP is considered to represent the highest sustainable rate of oxidative metabolism, whereas the curvature constant of the P – T_{lim} hyperbola (W') represents a fixed amount of work that can be performed above CP (19,25,26). The CP concept, therefore, describes an endogenous bioenergetic supply system that can be defined by a two-parameter model, namely, an oxidative component that is rate but not capacity-limited (CP) and a supplementary component that remains constant regardless of the rate at which it is expended (W') (19). Consistently low values of muscle phosphocreatine concentration ([PCr]) and pH

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Submitted for publication July 2011.
Accepted for publication October 2011.

0195-9131/12/4405-0966/0
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DOI: 10.1249/MSS.0b013e31823ea28a

have been reported at the limit of tolerance during CWR exercise $> CP$ (18,36). Theoretically, once the W' has been exhausted and the limit of tolerance attained, a reduction in work rate to below the CP is required (to enable a reconstitution of the W') if exercise is to be continued. The CP and W' can be estimated using a single all-out protocol where the W' is expended as a function of time until $W' = 0$, at which point the highest power output that can be elicited by maximal effort equals CP. The all-out test power profile is reliable (4), and the derived parameters show close agreement with the estimates from the conventional protocol (34,35) and can be used to accurately predict exercise tolerance (1,34).

Whereas the two-parameter CP model and associated physiological responses have been well described for severe CWR exercise, the profiles during intermittent exercise (i.e., severe “work” intervals separated by intervals of active or passive “recovery”) have been much less well examined. This is perhaps surprising given the widespread use of interval training by athletes and its potential applicability in the clinical setting (9,13,24). Morton and Billat (27) adapted the two-parameter CP model for intermittent exercise by including four independent variables (P and t during both work and recovery phases; i.e., P_w , t_w , P_r , and t_r , respectively) and found good fits to data collected during intermittent exercise. This suggests that the CP model is also appropriate for this form of exercise. However, there are conflicting findings as to whether CP and W' estimated from intermittent exercise are similar (and therefore have the same physiological equivalents) to those measured during continuous exercise (3,20,27).

The purpose of this investigation was to apply the CP model to better understand the physiological responses to intermittent exercise. Specifically, we hypothesized that, during exhaustive intermittent exercise in which severe exercise was interspersed with recovery periods in different exercise intensity domains (i.e., severe, heavy, moderate and light (i.e., 20 W)), the total work done above CP, and therefore the limit of tolerance, would be greatest for light, intermediate for moderate, and least for heavy recovery condition, whereas no W' recovery would occur in the severe recovery condition. We also hypothesized that changes in the overall rate of W' depletion during these intermittent exercise protocols would be reflected in changes to the rates at which $\dot{V}O_2$ and iEMG increased to the limit of tolerance.

METHODS

Subjects. Seven male subjects (mean $\pm$ SD: age = 26 ± 5 yr, height = 1.79 ± 0.06 m, body mass = 81 ± 6 kg) volunteered and gave written informed consent to participate in this study, which had been approved by the University of Exeter Research Ethics Committee. The subjects were all recreationally active and were familiar with the experimental procedures used in the study. On test days,

subjects were instructed to report to the laboratory in a rested state, having completed no strenuous exercise within the previous 24 h and having abstained from food, alcohol, and caffeine for the preceding 3 h. Testing was conducted at the same time of day (± 2 h) for each subject, and laboratory visits were separated by at least 48 h.

Experimental overview. All testing was completed at sea level in an air-conditioned laboratory at a temperature of 21°C. The subjects visited the laboratory on eight occasions during a 3-wk period to perform exercise tests on an electronically braked cycle ergometer (Lode Excalibur Sport, Groningen, The Netherlands). On the first visit, subjects completed a ramp incremental exercise test to determine $\dot{V}O_{2peak}$ and GET. During the second visit, subjects completed a familiarization 3-min all-out test, which was not used in data analysis. This test was repeated during the third laboratory visit and was used to estimate end-test power and work done above end-test power, which have been shown to closely approximate CP and W' , respectively (34,35). After these preliminary tests, subjects completed five cycling tests to exhaustion: on one occasion, the subjects completed a sustained severe-intensity exercise bout at CWR, and on the other occasions, subjects completed intermittent exercise during which 60 s of severe exercise was interspersed with 30 s of severe, heavy, moderate, or light exercise. These tests were presented to the subjects in randomized order. Before these tests, subjects' maximum iEMG (iEMG_{max}) of the right musculus vastus lateralis was determined. Strong verbal encouragement was provided during all of the tests that were required for this study.

Exercise tests. The ramp incremental exercise test consisted of 3 min of pedaling at 0 W, followed by a continuous ramped increase in work rate of $30 \text{ W} \cdot \text{min}^{-1}$ until the subject was unable to continue. The subjects cycled at a cadence of 80 rpm, and tests were terminated on volitional fatigue or when the required pedal rate could no longer be maintained (i.e., fell by >10 rpm). Peak power output (P_{peak}) was recorded, and $\dot{V}O_2$ data were averaged into 10-s bins. $\dot{V}O_{2peak}$ was defined as the highest 30-s rolling mean value recorded. The GET was determined from 10-s bin serial means from a cluster of measures including 1) the first disproportionate increase in carbon dioxide output ($\dot{V}CO_2$) from visual inspection of individual plots of $\dot{V}CO_2$ versus $\dot{V}O_2$, 2) an increase in $\dot{V}_E/\dot{V}O_2$ ($\dot{V}_E$, expiratory ventilation) with no increase in $\dot{V}_E/\dot{V}CO_2$, and 3) an increase in end-tidal O_2 tension with no fall in end-tidal CO_2 tension (39). The power output corresponding to the metabolic rate at which GET occurred (P_{GET}) was estimated with account taken of the mean response time of the $\dot{V}O_2$ response, which was assumed to approximate 40 s.

The 3-min all-out test began with 3 min of “unloaded” (i.e., 0 W) baseline pedaling at 80 rpm, followed by a 3-min all-out effort (4). Subjects were asked to accelerate their cadence to ~ 110 – 120 rpm during the last 5 s of the baseline period. The resistance on the pedals during the 3-min all-out effort was set using the linear mode of the Lode

Excalibur Sport ergometer such that the subject would attain the power output halfway between P_{GET} and P_{peak} (see above) on reaching their preferred cadence (linear factor = power/preferred cadence²). The preferred cadence was 80 rpm for all subjects in this study. To prevent pacing, subjects were not informed of the elapsed time. To ensure an all-out effort, subjects were instructed to attain their peak power output as quickly as possible from the start of the test and to maintain their cadence as high as possible at all times throughout. The $\dot{V}O_{2peak}$ during the 3-min all-out test was defined as the highest 15-s serial mean value recorded during the bout. The CP was estimated as the mean power output during the final 30 s of the test (EP) and the W' was estimated as the power-time integral above EP using commercially available software (Microsoft Excel, 2007; Microsoft Corp., Redmond, WA).

The power output that would be predicted to lead to exhaustion in 6 min (P_6) was then calculated from the following equation:

$$P_6 = (W'/t_{desired}) + CP \quad [1]$$

where P is the power output, $t_{desired}$ is the desired time-to-exhaustion (i.e., 360 s), CP is the critical power (estimated as EP in the all-out test), and W' is the finite work capacity > CP in joules (estimated as work done above EP).

Subjects returned to the laboratory on five occasions to perform the CWR and intermittent exercise tests (Fig. 1). Subjects cycled at a cadence of 80 rpm during all tests. The severe CWR bout (S-CWR) began with 3 min of 0-W baseline cycling after which work rate was abruptly increased to P_6 . Subjects were instructed to continue for as long as possible, and the test was terminated when cadence fell by >10 rpm. For the intermittent exercise bouts, subjects completed four different exercise protocols in which 60 s of exercise at a severe work rate was separated by 30 s of exercise at lower ("recovery") work rates (Fig. 1). In each case, the test began with the same baseline cycling period

as S-CWR and was continued until cadence fell by >10 rpm. All intermittent exercise tests included "work" intervals performed at a severe work rate of P_{6+} (i.e., $P_6 + 50\% \Delta_2$, where $\Delta_2 = P_6 - CP$). However, the work rates during the "recovery" intervals were varied for the four conditions as subjects cycled at either a lower severe work rate of P_{6-} (i.e., $P_6 - 50\% \Delta_2$), a heavy-intensity work rate halfway between P_{GET} and CP (i.e., at $P_{GET} + 50\% \Delta_1$, where Δ_1 = the range of work rates between P_{GET} and CP), a moderate-intensity work rate (90% P_{GET}), or a light work rate of 20 W. These four intermittent exercise bouts were designated as S-S, S-H, S-M, and S-L, respectively. Subjects were not informed of the work rates, the elapsed time, the expected time-to-exhaustion, or their performance in any of the tests until the entire study had been completed.

Before the CWR and intermittent exercise tests, the right leg was shaved and cleaned with alcohol around the belly of the musculus vastus lateralis, and graphite snap electrodes (Unilect, Unomedical Ltd., Deeside, UK) were adhered to the prepared area in a bipolar arrangement (interelectrode distance = 40 mm). A ground electrode was positioned on the musculus rectus femoris equidistant from the active electrodes. To secure electrodes and wires in place and to minimize movement during cycling, an elastic bandage was wrapped around the subject's leg. Pen marks were made around the electrodes to enable reproduction of the placement in subsequent tests. For the assessment of maximal voluntary contraction (MVC) and associated neuromuscular activity (EMG), subjects sat on an isokinetic dynamometer (Biodex Isokinetic; Biodex Medical Ltd., Shirley, New York) with their hips, thighs, and torso strapped securely to the seat and their right lower leg strapped securely to the movement arm. The seat was positioned such that the subject's hip angle was 100°, and the movement arm was set such that their knee angle was fixed at 120° (anatomical joint angle; i.e., full extension = 180°). From this position, subjects performed three maximal isometric knee extension contractions of 5 s in duration separated by 60 s of recovery. The contraction that elicited the highest peak torque was considered the MVC. Once this testing was completed, 15 min of rest was allowed before beginning the exhaustive cycling bouts.

Measurements. During all cycling tests, pulmonary gas exchange was measured breath-by-breath with subjects wearing a nose clip and breathing through a low dead space, low-resistance mouthpiece, and an impeller turbine assembly (Jaeger Triple V; Erich Jaeger GmbH, Hoechberg, Germany). The inspired and expired gas volumes and gas concentration signals were continuously sampled at 100 Hz, the latter using paramagnetic (O_2) and infrared (CO_2) analyzers (Jaeger Oxycon Pro; Erich Jaeger GmbH) via a capillary line connected to the mouthpiece. These analyzers were calibrated before each test with gases of known concentration, and the turbine volume transducer was calibrated using a 3-L syringe (Hans Rudolph, Inc., Kansas City, MO). The volume and concentration signals were time aligned by accounting for the delay in capillary gas transit and analyzer rise time

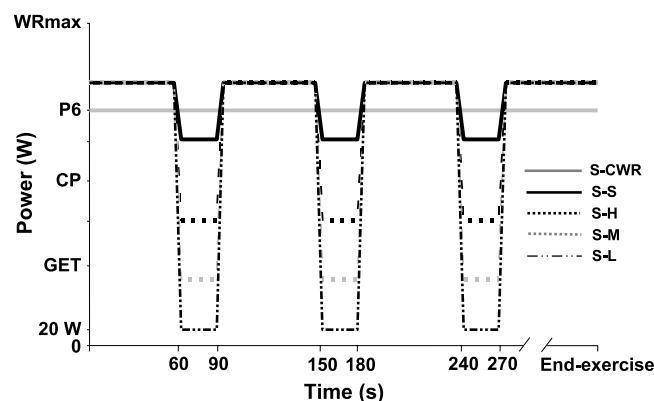


FIGURE 1—Schematic of the experimental protocol. Subjects completed S-CWR exercise bout and four intermittent protocols in which 60 s of severe exercise was interspersed with 30 s of exercise at a lower work rate in different exercise intensity domains (severe, heavy, moderate, and light). See text for further details.

relative to the volume signal. Oxygen uptake, $\dot{V}CO_2$, and $\dot{V}_E$ were displayed breath-by-breath. HR was measured every 5 s during all tests using short-range radiotelemetry (Polar S610; Polar Electro Oy, Kempele, Finland).

During the CWR and intermittent exercise tests, EMG activity was recorded using a ME3000PB Muscle Tester (Mega Electronics Ltd., Kuopio, Finland) at a sampling frequency of 1000 Hz. The bipolar signal was amplified (amplifier input impedance >1 M Ω), and data were collected online in raw form and stored on a personal computer using MegaWin software (Mega Electronics). The raw EMG data were subsequently exported as an ASCII file and digitally filtered using LabView 8.2 (National Instruments, Newbury, UK). Initially, the signals were filtered with a 20-Hz high-pass, second-order Butterworth filter to remove contamination from movement artifacts. A linear envelope was then produced via rectification (transformation of all signals to absolute values by converting negative values to positive values) and low pass (500 Hz) filtering to smooth high-frequency peaks (e.g., signals in the external environment detected by the electrodes). This process improves the ability to detect relevant aspects of the signal.

During the CWR and intermittent exercise tests, a blood sample from a fingertip was collected into a capillary tube during the last 20 s of baseline cycling and within 20 s of the termination of exercise. These samples were immediately analyzed to determine blood [lactate] (YSI 1500; Yellow Springs Instruments, Yellow Springs, OH).

Data analysis procedures. The total work done ($W = PT_{lim}$) before the limit of tolerance and the average power output during each of the exercise tests were calculated. Subsequently, mathematically equivalent linear transformations of the hyperbolic P - T_{lim} relationship were used to determine two sets of CP and W' estimates according to the following equations (i.e., the W - T_{lim} model and $1/T_{lim}$ model, respectively):

$$W = (CPT_{lim}) + W' \quad [2]$$

$$P = W' (1/T_{lim}) + CP \quad [3]$$

where P indicates the mean power output during the entire trial. These parameters were also calculated using the version of the two-parameter model that has been modified for intermittent exercise (27):

$$t = n(t_w + t_r) + [W' - n\{(P_w - CP)t_w - (CP - P_r)t_r\}]/(P_w - CP) \quad [4]$$

where t is total endurance time, n is the number of completed work-recovery cycles, t_w and t_r are the durations, and P_w and P_r the power outputs of the work and recovery intervals, respectively. P and t over all of the repeated work-recovery cycles were cumulated, and then the CP and W' parameters were derived using nonlinear regression.

To assess the work done above CP ($W = PT_{lim}$; expressed in kilojoules) during the CWR and intermittent

exercise bouts, we used the following equations for S-CWR (equation 5a), S-S (equation 5b), and S-H, S-M, and S-L (equation 5c):

$$W_{>CP} = [(P_6 T_{lim}) - (CP T_{lim})]/1000 \quad [5a]$$

$$W_{>CP} = [(P_{6+ t_w}) + (P_{6- t_r}) - (CP T_{lim})]/1000 \quad [5b]$$

$$W_{>CP} = [(P_{6+ t_w}) - (CP t_w)]/1000 \quad [5c]$$

where T_{lim} is the time-to-exhaustion, t_w is the cumulative time spent during intermittent work intervals, and t_r is the cumulative time spent during intermittent recovery intervals.

The breath-by-breath $\dot{V}O_2$ data collected during the CWR and intermittent exercise tests were initially examined to exclude errant breaths caused by coughing, swallowing, sighing, etc., and those values lying >4 SD from the local mean (five-point rolling mean) were removed. The remaining data were subsequently linearly interpolated to provide second-by-second values. $\dot{V}O_{2baseline}$ was defined as the mean $\dot{V}O_2$ measured during the final 90 s of cycling before the onset of exercise, whereas the end-exercise $\dot{V}O_2$ was defined as the mean $\dot{V}O_2$ measured during the final 15 s of exercise. To estimate the total O_2 consumed (L), we used the second-by-second $\dot{V}O_2$ data to calculate the $\dot{V}O_2$ -time integral above baseline for each experimental condition up to 180 s, 300 s (the longest time that all subjects were able to reach for all conditions), and the end of exercise. Baseline and end-exercise HR were defined as the mean HR measured during the final 15 s of baseline cycling and the final 15 s of each exercise bout, respectively.

Because of the intermittent nature of the exercise tests used in this study, we did not attempt to formally characterize $\dot{V}O_2$ kinetics via conventional modeling procedures. However, we did obtain a broad characterization of “delayed-onset” response dynamics by calculating the slope of the $\dot{V}O_2$ response using linear regression analysis. The first 60 s of data were deleted to remove the influence of the fundamental response phase and, thereafter, $\dot{V}O_2$ values at 90-s intervals (i.e., the average $\dot{V}O_2$ during the 15 s before completion of each work cycle during intermittent exercise) were determined up to the last completed cycle (and to end-exercise for S-CWR) and fit using the following equation:

$$\dot{V}O_2 = ax + b \quad [6]$$

where x is the time, a is the slope, and b is the y -intercept.

The average iEMG was calculated for 1-s intervals during the MVC and for 10-s intervals throughout both the baseline and exercise periods of the CWR and intermittent exercise tests. The iEMG_{max} in the MVC was considered as the highest 1-s value, and iEMG data collected during the subsequent cycling bout were normalized to this value. The baseline iEMG was defined as the average iEMG during the 90 s preceding the onset of exercise and the end-exercise iEMG

was defined as the average iEMG during the final 10 s of exercise. We then calculated the slope of the iEMG response during exercise using a similar procedure to that already described for calculating the $\dot{V}O_2$ slope (see above). However, iEMG was averaged during the final 10 s of each work interval and the first work interval was not excluded from the fitting window.

Statistical analysis. A one-way repeated-measures ANOVA was used to determine the effects on the relevant physiological and performance variables elicited by the work rate functions for the CWR and intermittent exercise tests and the comparison of $\dot{V}O_2$ peak between all exercise tests. Where the analysis revealed a significant difference, individual paired *t*-tests were used with LSD to determine the origin of such effects. All data are presented as mean $\pm$ SD. Statistical significance was accepted when $P < 0.05$.

RESULTS

The $\dot{V}O_{2\text{peak}}$ measured during the ramp incremental test was $4.10 \pm 0.78 \text{ L} \cdot \text{min}^{-1}$. P_{peak} and P_{GET} were 371 ± 62 and $105 \pm 15 \text{ W}$, respectively. The CP and W' estimated from the 3-min all-out test were $241 \pm 59 \text{ W}$ and $21.1 \pm 4.7 \text{ kJ}$, respectively. The $\dot{V}O_{2\text{peak}}$ measured during the 3-min all-out test was $4.06 \pm 0.89 \text{ L} \cdot \text{min}^{-1}$, which was not significantly different from the $\dot{V}O_{2\text{peak}}$ measured during the ramp incremental test ($P > 0.05$). The CWR exercise bout (i.e., P_6) was completed at $300 \pm 60 \text{ W}$, and the work interval for the intermittent exercise bouts (i.e., P_{6+}) was $329 \pm 61 \text{ W}$. The work rates applied during the recovery intervals were 270 ± 59 , 173 ± 35 , 95 ± 13 , and $20 \pm 0 \text{ W}$ for S-S, S-H, S-M, and S-L, respectively. The mean work rates were therefore 309 ± 60 , 277 ± 52 , 251 ± 44 , and $226 \pm 41 \text{ W}$ for S-S, S-H, S-M, and S-L, respectively. The mean work rates for S-S and S-H were significantly higher than the CP ($P < 0.05$), whereas the mean work rates for S-M and S-L were not significantly different from the CP.

Table 1 presents physiological variables and T_{lim} values during the CWR and intermittent severe cycling bouts. T_{lim} was significantly different in all cases. Specifically, T_{lim} was $384 \pm 48 \text{ s}$ for S-CWR, was lower by $\sim 15\%$ for S-S, and was higher by $\sim 47\%$ for S-H, by $\sim 100\%$ for S-M, and by $\sim 219\%$ for S-L. The $W_{>\text{CP}}$ was not significantly different between S-CWR and S-S ($\sim 23 \text{ kJ}$) or from the subjects' W' estimated from the 3-min all-out test ($P > 0.05$). However, the $W_{>\text{CP}}$ became progressively greater as the recovery work rate was reduced from S-H to S-M to S-L (Table 1). The end-exercise $\dot{V}O_2$ from the CWR and intermittent exercise tests were not significantly different from one another or from the $\dot{V}O_{2\text{peak}}$ values measured during the ramp incremental test or the 3-min all-out test, with the exception that end-exercise $\dot{V}O_2$ was significantly lower for S-L compared with S-CWR (Table 1). The group mean $\dot{V}O_2$ response profiles are depicted in Figure 2.

A significant reduction in the $\dot{V}O_2$ response slope was present for S-H, S-M, and S-L compared with both S-CWR and S-S. There was no significant difference for total O_2 consumed during the first 180 or 300 s of exercise for S-S compared with S-CWR; however, the values for S-H, S-M, and S-L became progressively smaller in both cases (Table 1). A similar pattern was present when O_2 consumed was calculated to end-exercise.

Blood [lactate] and iEMG measurements are also presented in Table 1, and group mean iEMG response profiles are depicted in Figure 3. There was no significant difference in baseline or end-exercise blood [lactate] or baseline iEMG between conditions. End-exercise iEMG was significantly less for S-M compared with S-CWR, S-S, and S-H and for S-L compared with all other conditions; there was no significant difference in end-exercise iEMG between S-CWR, S-S, and S-H. The iEMG slopes showed a progressive decline for S-H, S-M, and S-L (Table 1).

The estimates of CP and W' derived from the work-time model, the 1/time model, and the intermittent model along with the estimates from the 3-min all-out test are provided

TABLE 1. Selected physiological variables and limit of tolerance during CWR and intermittent severe-intensity cycling.

	S-CWR	S-S	S-H	S-M	S-L
Baseline HR (beats·min ⁻¹)	85 $\pm$ 14	87 $\pm$ 13	88 $\pm$ 9	87 $\pm$ 12	85 $\pm$ 10
End-exercise HR (beats·min ⁻¹)	182 $\pm$ 10	178 $\pm$ 8	181 $\pm$ 8	177 $\pm$ 9	175 $\pm$ 12
Baseline $\dot{V}O_2$ (L·min ⁻¹)	1.08 $\pm$ 0.13	1.06 $\pm$ 0.16	1.09 $\pm$ 0.12	1.06 $\pm$ 0.13	1.09 $\pm$ 0.14
End-exercise $\dot{V}O_2$ (L·min ⁻¹)	4.25 $\pm$ 0.86	4.18 $\pm$ 0.78	4.10 $\pm$ 0.59	4.04 $\pm$ 0.60	3.89 $\pm$ 0.77*
Slope $\dot{V}O_2$ (L·min ⁻²)	0.16 $\pm$ 0.03	0.19 $\pm$ 0.11	0.09 $\pm$ 0.02**	0.09 $\pm$ 0.01**	0.07 $\pm$ 0.02***
Total $\dot{V}O_2$ over 180 s (L)	9.4 $\pm$ 1.8	9.5 $\pm$ 1.8	9.1 $\pm$ 1.5***	8.9 $\pm$ 1.5****	8.4 $\pm$ 1.6*****
Total $\dot{V}O_2$ over 300 s (L)	17.6 $\pm$ 3.4	17.6 $\pm$ 3.5	16.5 $\pm$ 2.7**	15.7 $\pm$ 2.5****	14.8 $\pm$ 2.7*****
Total $\dot{V}O_2$ end (L)	23.9 $\pm$ 6.5	19.7 $\pm$ 5.9*	33.9 $\pm$ 2.3**	46.1 $\pm$ 23.9****	71.8 $\pm$ 39.8*****
Baseline blood [lactate] (mM)	0.9 $\pm$ 0.2	0.9 $\pm$ 0.1	0.9 $\pm$ 0.2	0.9 $\pm$ 0.2	1.1 $\pm$ 0.3
End-exercise blood [lactate] (mM)	8.4 $\pm$ 1.4	9.2 $\pm$ 1.8	9.5 $\pm$ 2.3	9.0 $\pm$ 1.4	8.9 $\pm$ 1.3
Baseline iEMG (% iEMG _{max})	8 $\pm$ 4	8 $\pm$ 4	8 $\pm$ 4	8 $\pm$ 4	8 $\pm$ 3
End-exercise iEMG (% iEMG _{max})	44 $\pm$ 11	40 $\pm$ 11	39 $\pm$ 8	34 $\pm$ 10****	34 $\pm$ 10****
Slope iEMG (% iEMG _{max} per minute)	2.1 $\pm$ 1.0	1.8 $\pm$ 1.4	0.8 $\pm$ 0.6*	0.2 $\pm$ 0.4*	0.1 $\pm$ 0.2****
T_{lim} (s)	384 $\pm$ 48	323 $\pm$ 29*	557 $\pm$ 90**	759 $\pm$ 243****	1224 $\pm$ 497*****
$W_{>\text{CP}}$ (kJ)	22.9 $\pm$ 7.4	22.8 $\pm$ 5.0	33.4 $\pm$ 8.2**	45.3 $\pm$ 16.3****	72.3 $\pm$ 27.2*****

* Significantly different from S-CWR ($P < 0.05$).

** Significantly different from S-CWR and S-S ($P < 0.05$).

*** Significantly different from S-S ($P < 0.05$).

**** Significantly different from S-CWR, S-S, and S-H ($P < 0.05$).

***** Significantly different from S-CWR, S-S, S-H, and S-M ($P < 0.05$).

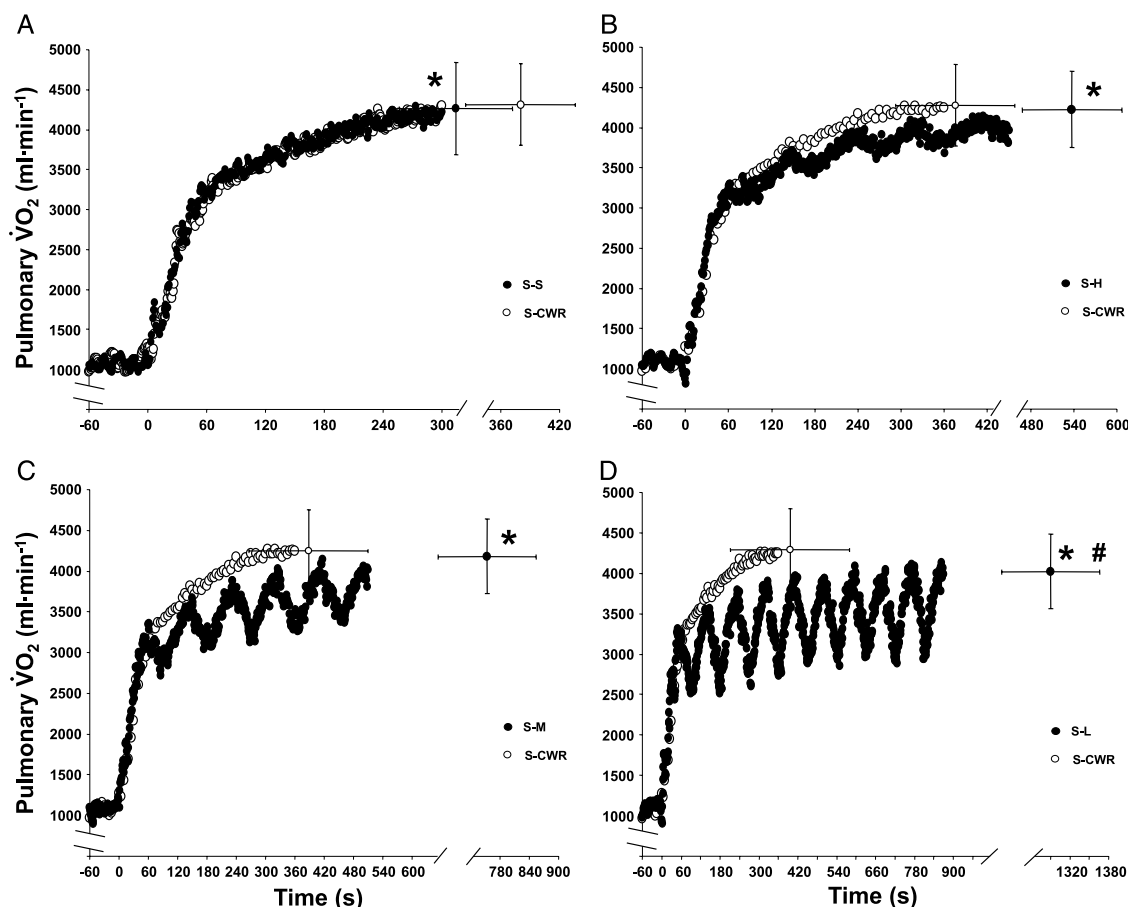


FIGURE 2—Group mean pulmonary $\dot{V}O_2$ response to S-CWR (open circles) compared with intermittent exercise (closed circles) for S-S (A), S-H (B), S-M (C), and S-L (D). The mean $\pm$ SD values at end-exercise are also shown. * T_{lim} is significantly different from S-CWR ($P < 0.05$). #End-exercise $\dot{V}O_2$ is significantly lower than S-CWR ($P < 0.05$).

for each subject in Table 2. The R^2 values ranged from 0.99 to 1.00, from 0.99 to 1.00, and from 0.95 to 1.00 for the work-time model, the 1/time model, and the intermittent model, respectively. There were no significant differences in the CP or W' estimates between the three different procedures. However, the values for CP and W' were significantly different compared to those derived using the 3-min all-out test (Table 2).

DISCUSSION

The principal original finding of this investigation was that the $W_{>CP}$ and limit of tolerance during intermittent exercise differed significantly when the recovery intervals were positioned in different intensity domains ($<CP$). During CWR or intermittent exercise performed exclusively in the severe domain, the $W_{>CP}$ was not significantly different from the subjects' W' , as estimated from the 3-min all-out test (34). This confirms and extends the notion that, in this domain, the limit of tolerance is reached when subjects expend the finite work capacity available above CP, i.e., the W' (11). When intermittent exercise involving work bouts in the severe domain and recovery intervals in lower-intensity domains was performed, however, the limit of

tolerance (and the $W_{>CP}$) was extended, being greatest for S-L, intermediate for S-M, and least for S-H. The slopes describing the increases in $\dot{V}O_2$ and iEMG with time during intermittent exercise were reduced when the recovery intervals were completed at progressively lower intensities. These findings support the recent suggestion that depletion of the W' is linked to the $\dot{V}O_2$ "slow component" that is observed during S-CWR (5,38). It is feasible that both phenomena (i.e., the W' and the $\dot{V}O_2$ slow component) reflect muscle fatigue development (5,28,36,38).

There was no significant difference in $W_{>CP}$ between S-CWR and S-S (~ 22.8 kJ) and this value was also not significantly different from the W' as estimated from the 3-min all-out test (4,34,35). This is consistent with previous studies which have reported that $W_{>CP}$ equals W' at the limit of tolerance during severe exercise (11,14,30,42). The highest $\dot{V}O_2$ attained was not significantly different between S-CWR and S-S and was not significantly different from the $\dot{V}O_{2peak}$ measured during either the ramp incremental test or the 3-min all-out test. The attainment of a similar $\dot{V}O_{2peak}$ with different work-rate impositions (i.e., all-out sprint, ramp incremental, and CWR) within the severe domain is also consistent with previous studies (4,7,11,12,14,42).

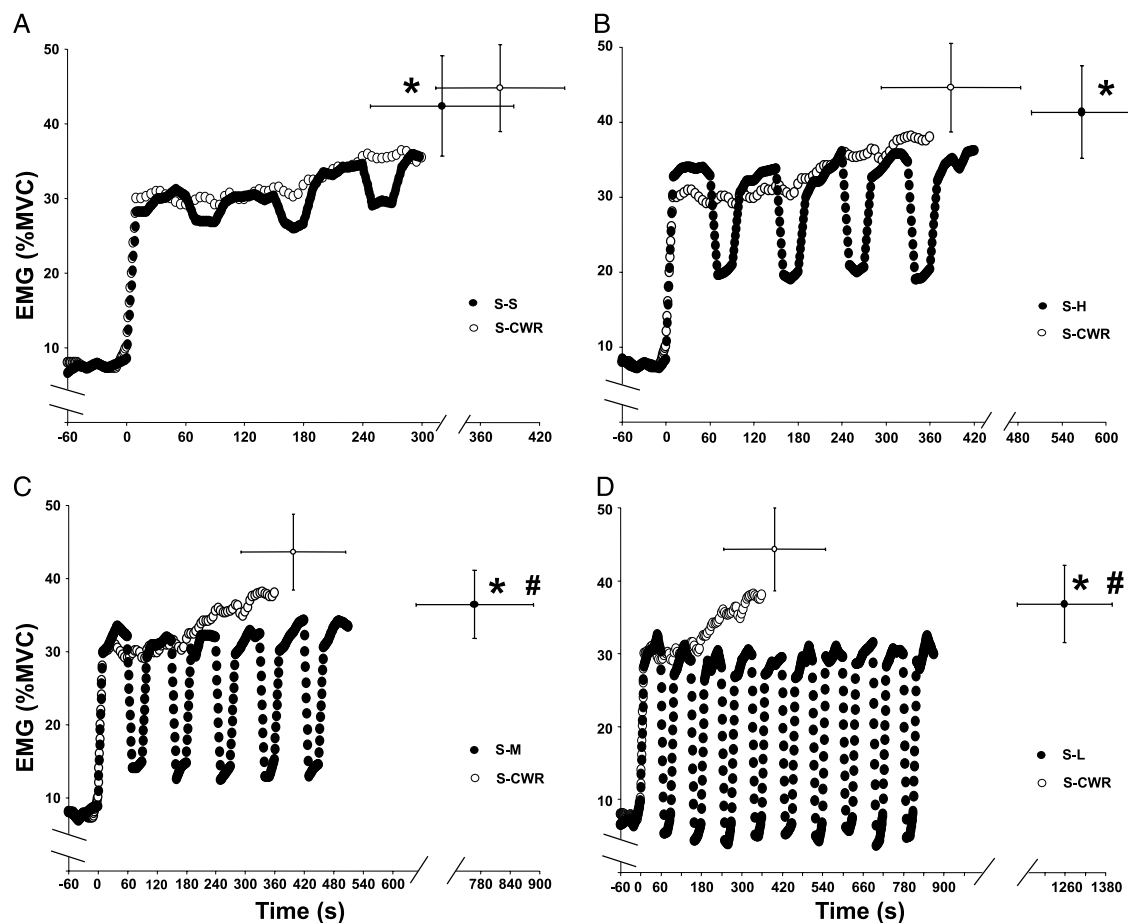


FIGURE 3—Group mean iEMG response to S-CWR (open circles) compared with intermittent-work exercise (closed circles) for S-S (A), S-H (B), S-M (C), and S-L (D). The mean $\pm$ SD values at end-exercise are also shown. * T_{lim} is significantly different from S-CWR ($P < 0.05$). #End-exercise iEMG is significantly lower than S-CWR ($P < 0.05$).

To investigate the influence of the CP and W' on performance during intermittent exercise, we asked subjects to perform intermittent exercise protocols (using a “60-s work, 30-s recovery” duty cycle) until the limit of tolerance. The intensity of the recovery intervals was selected to span the exercise-intensity domains, i.e., S-S, S-H, S-M, and S-L. In the S-S condition, the recovery interval was itself $>CP$, and thus theoretically, there was no opportunity for W' to be reconstituted between the work bouts (11). Accordingly, the $W_{>CP}$ in S-CWR and S-S was not different, and the limit of tolerance during S-S could be accurately predicted by rear-

ranging equation 1. Relative to S-CWR and S-S, the limit of tolerance became progressively longer and the $W_{>CP}$ became progressively greater when the intensity of the recovery interval was reduced in the S-H, S-M, and S-L conditions. These findings can be explained by considering that some of the W' that is expended during a severe work bout may be reconstituted during a subsequent recovery period provided that the recovery work rate is $<CP$.

The contention that the W' is expended during work bouts $>CP$ and reconstituted during recovery intervals $<CP$ may be understood with reference to the recent study of

TABLE 2. Parameters of the power-duration relationship derived from the 3-min all-out test, the intermittent CP model, and two conventional two-parameter model equations.

Subject	Critical Power (W)				W (kJ)				R^2		
	3 min	Intermittent	Work-Time	1/Time	3 min	Intermittent	Work-Time	1/Time	Intermittent	Work-Time	1/Time
1	211	182	188	188	28.0	43.5	39.9	39.9	0.98	0.99	0.99
2	220	195	196	196	25.4	36.5	36.0	36.0	0.99	0.99	0.99
3	213	182	183	183	22.3	37.9	36.8	36.8	1.00	1.00	1.00
4	359	279	281	281	21.5	63.9	62.9	62.9	0.98	0.99	0.99
5	277	224	224	224	18.2	39.9	40.2	40.2	1.00	1.00	1.00
6	187	151	151	151	17.8	33.0	32.9	32.9	0.99	1.00	1.00
7	221	191	195	194	14.3	22.8	20.8	21.7	0.95	0.99	0.99
Mean	241*	200	203	202	21.1*	39.6	38.5	38.6	0.98	0.99	0.99
SD	59	41	41	41	4.7	12.5	12.6	12.4	0.02	0.00	0.00

* Significantly different from other three models ($P < 0.05$).

Jones et al. (18). Using ^{31}P magnetic resonance spectroscopy, these authors showed that, when CWR exercise was performed slightly above CP, the intramuscular [PCr] and pH continued to decrease, and inorganic phosphate concentration ($[\text{P}_i]$) continued to increase, until the limit of tolerance was reached. During exercise performed just below the CP, however, stable values for [PCr], pH, and $[\text{P}_i]$ were attained within 3 min of the start of exercise (18). From these results, it may be hypothesized that the recovery intervals during intermittent exercise allow some of the fatigue-related substrates to be resynthesized (e.g., PCr) and for fatigue-related metabolites to be cleared from the muscle (e.g., H^+), thereby delaying the attainment of a “limiting” intramuscular environment (18,36). This suggestion is consistent with the study of Coats et al. (6). These authors asked subjects to perform S-CWR to the limit of tolerance (attained in around 6 min) and then reduced the work rate to 80% GET, 90% CP, or 110% CP. The results indicated that replenishment of the W' following the limit of tolerance necessitated that work rate be reduced <CP (6).

The extent to which the W' was reconstituted in the recovery intervals during intermittent exercise can be calculated using the group mean data, as follows. Given a severe work rate of 329 W and a CP of 241 W, the $W_{>\text{CP}}$ in one 60-s bout of severe exercise is $(329 - 241) \text{ W} \times 60 = 5.3 \text{ kJ}$. Taking the S-H condition as an example, the subjects completed 6.18 work-recovery cycles on average, during which they accumulated $\sim 33.2 \text{ kJ}$ of work >CP ($6 \times 5.3 \text{ kJ} = 31.7 \text{ kJ}$ plus $(329 - 241) \times 17 \text{ s} = 1.5 \text{ kJ}$ during the final incomplete stage). The W' reconstituted during the six 30-s recovery intervals between severe work bouts can be calculated to be 2.01 kJ (i.e., $(\text{total } W_{>\text{CP}} - W') / \text{the number of recovery cycles completed}$; $(33.2 - 21.1 \text{ kJ})/6 = 2.05 \text{ kJ}$). During 6.18 work-recovery cycles ($T_{\text{lim}}/90 \text{ s}$), this would mean an additional 12.1 kJ of $W_{>\text{CP}}$ or a total $W_{>\text{CP}}$ (including the initial W' of 21.1 kJ) before the limit of tolerance of approximately 33.2 kJ. Using the same procedures for the S-M condition, the W' reconstituted in each recovery interval is 3.1 kJ, and the total $W_{>\text{CP}}$ is $\sim 45 \text{ kJ}$, whereas for the S-L condition, the W' reconstituted in the recovery interval is 4.0 kJ and the total $W_{>\text{CP}}$ is $\sim 72 \text{ kJ}$. These hypothesized relationships between W' depletion and reconstitution with different recovery intensities are illustrated in Figure 4. It is clear that the reconstitution of the W' was more complete when the absolute recovery work rate between the severe work bouts was lower, thereby enabling a greater $W_{>\text{CP}}$ and extending the limit of tolerance. It may be speculated that lower recovery work rates enabled a more rapid reconstitution of the W' because a larger fraction of the muscle fibers that were metabolically active during the severe work bouts were no longer contributing to power production during the recovery interval.

Interestingly, there was a significant reduction in the slope of the relationship between $\dot{V}\text{O}_2$ and time in the conditions where the recovery work rate was <CP (S-H, S-M,

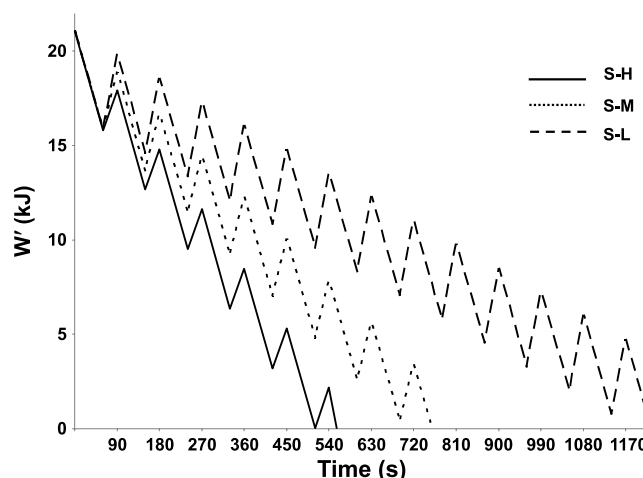


FIGURE 4—Schematic illustration of the depletion of W' during >CP work bouts and the reconstitution of W' during subsequent <CP recovery intervals. Notice that the reconstitution of W' is greater when the recovery intervals are completed at lower intensities, delaying the complete depletion of W' and enabling a longer T_{lim} . For simplicity, the model assumes that W' is reconstituted linearly during the recovery intervals and that the rate of reconstitution is the same irrespective of the magnitude of W' depletion.

and S-L) compared with >CP CWR or intermittent exercise (S-CWR and S-S, respectively). During S-CWR, $\dot{V}\text{O}_2$ rises inexorably until $\dot{V}\text{O}_{2\text{peak}}$ is attained (14,30,42). In this exercise domain, the development of the $\dot{V}\text{O}_2$ slow component and the depletion of the W' occur concomitantly (5,15,28,38). It has been proposed that these characteristic features of the physiological response to severe exercise are mechanistically linked and that both are sequelae of muscle fatigue development (5,28,38). Historically, the W' has been considered to represent a finite amount of energy in stored O_2 , the high-energy phosphates, and a source related to anaerobic glycolysis that may be expended above the CP (25,26). An alternative perspective, however, is that the W' is related to the accumulation or depletion of one or more metabolites or substrates that are linked to the process of muscle fatigue until some “critical” concentration is attained, beyond which the same work rate cannot be tolerated (6,11,17,18). Consistent with this, it has been shown that the limit of tolerance during a series of S-CWR bouts was associated with the attainment of consistently low values of muscle [PCr] and pH and a consistently high value of muscle $[\text{P}_i]$ (36). The present study shows that, compared with S-CWR and S-S, the total O_2 consumed during the first 180 and 300 s of exercise was significantly and progressively reduced in S-H, S-M, and S-L. This blunting of the rate at which $\dot{V}\text{O}_2$ increased with time in these conditions resulted in a delayed attainment of $\dot{V}\text{O}_{2\text{peak}}$ and an extension of the limit of tolerance. The simultaneous blunting of W' expenditure and the $\dot{V}\text{O}_2$ -time slope (reflecting a loss of oxidative metabolic efficiency) during intermittent exercise is consistent with the proposed mechanistic relationship between the W' and the $\dot{V}\text{O}_2$ slow component during S-CWR (5,15,36,38).

The finding that the $\dot{V}O_2$ response profile can be modulated during intermittent exercise is consistent with the study of Turner et al. (33). These authors examined pulmonary gas exchange and blood lactate dynamics during intermittent exercise protocols in which cycling work intervals at 120% of peak work rate were separated by recovery intervals at 20 W at different duty cycles (10:20 s, 30:60 s, 60:120 s, and 90:180 s). The 10:20-s duty cycle produced a physiological response profile that was akin to that observed during moderate-intensity CWR exercise (i.e., low blood [lactate] and first-order $\dot{V}O_2$ kinetics); the 30:60-s duty cycle produced responses that were similar to those observed during heavy-intensity CWR exercise (elevated but stable blood [lactate] and evidence of a supplementary but low-amplitude $\dot{V}O_2$ slow component), and the 60:120-s and 90:180-s duty cycles resulted in responses that were consistent with S-CWR (high blood [lactate] and a $\dot{V}O_2$ slow component projecting to $\dot{V}O_{2peak}$). Only two of six subjects were able to complete 30 min of exercise in the 60:120-s condition, and none of the subjects could complete 30 min of exercise in the 90:180-s condition. Collectively, the present study and that of Turner et al. (33) indicate that the trajectory of $\dot{V}O_2$ toward $\dot{V}O_{2peak}$ reflects the development of muscular fatigue and is an important determinant of exercise tolerance during intermittent exercise.

Similar to the reduction in the $\dot{V}O_2$ -time slopes with decrements in recovery work rate <CP, the iEMG slopes declined in S-H, S-M, and S-L, implying that neuromuscular activity (involving recruitment of additional motor units and/or increased rate coding) was attenuated with time in comparison to S-CWR and S-S. The $\dot{V}O_2$ slow component arises predominantly from within the contracting muscles (21,29) and has been suggested to be related, at least in part, to the recruitment of fast-twitch fibers during high-intensity exercise (2,15). These fibers are considered to be less fatigue resistant and less efficient than their slow-twitch counterparts, and their recruitment would be expected to increase energy turnover and lead to a concomitant and progressive increase in $\dot{V}O_2$ (2,8,22,23,38). Collectively, the results indicate that the recovery intervals between the severe work bouts during intermittent exercise reduce the rate of fatigue development (consistent with the calculable reconstitution of the W') and, in so doing, blunt both the neuromuscular activity required to meet the demands of the exercise task and the associated rise in $\dot{V}O_2$ toward the limit of tolerance.

The end-exercise iEMG was significantly lower for S-M and S-L compared with the other conditions, and the end-exercise $\dot{V}O_2$ was significantly lower for S-L compared with the other conditions. The mean work rates sustained during S-S (309 ± 60 W) and S-H (277 ± 52 W) were significantly higher than the CP estimated from the 3-min all-out test (241 ± 59 W). The mean work rate sustained during S-M (251 ± 44 W), however, was only ~ 10 W higher than the estimated CP, and the mean work rate sustained during S-L (226 ± 41 W) was ~ 15 W lower than the estimated CP. That the end-exercise $\dot{V}O_2$ for S-L was lower than the

$\dot{V}O_{2peak}$ is consistent with exercise performed in the heavy-intensity domain (19,30,42). Moreover, the lower end-exercise iEMG at work rates $\leq$ CP may suggest exercise intensity domain-specific differences in fiber recruitment or rate coding, which may be linked to differences in fatigue mechanisms <CP compared with >CP (5,18,30).

This is the first study to apply the intermittent CP model of Morton and Billat (27) to cycling exercise. Using the data collected during the CWR and four intermittent exercise protocols, there were no significant differences in either the CP or W' parameter estimates when the work-time model, the 1/time model, and the intermittent model (27) (see Jones et al. (19) for review) were applied. It is important to note, however, that the work-time and 1/time models (equations 2 and 3) provide meaningful parameters only when applied to continuous exercise >CP. In the present study, the total work done and mean power output during intermittent exercise were modeled using these equations to illustrate the goodness of fit and absence of random error in prediction trial data. It is acknowledged that equations 2 and 3 do not account for W' recovery and make the erroneous assumption that $P > CP$ when applied to intermittent exercise. Although equation 4 is an improvement on equations 2 and 3 in that it accounts for linear W' recovery, the precise recovery kinetics of W' during intermittent exercise warrant further study (10).

The CP and W' parameter estimates (equations 2–4) were significantly lower and higher, respectively, compared with the values derived from the 3-min all-out test—which has been shown to provide close estimates of the conventionally established CP and W' parameters during continuous exercise (34,35). These results are similar to those of Morton and Billat (27) who reported that the two-parameter intermittent CP model (equation 4) resulted in a lower CP and a higher W' during intermittent compared with continuous running. This is consistent with the notion that the W' is fixed at the onset of S-CWR but may be reconstituted during intermittent exercise where the recovery intervals are <CP. It is unclear why CP is lower during intermittent compared with continuous exercise. It is possible that this may be related to the apparent reciprocity of CP and W' estimates; for example, interventions such as endurance training and hyperoxia which increase the CP also tend to reduce the W' (35,36). Alternatively, it is possible that CP may be lower during intermittent compared with continuous exercise as a direct consequence of the more substantial nonoxidative contribution to energy turnover during repeated short bouts of high-intensity exercise (19).

In conclusion, the results of this study indicate that physiological responses and exercise tolerance during intermittent exercise can be understood in terms of the CP concept. Recovery intervals between severe work bouts enable the finite W' to be restored, with the magnitude of this reconstitution being related to the intensity and duration of the recovery interval. From a practical perspective, prior knowledge of the CP and W' for intermittent exercise

could be used to predict the limit of tolerance (37) and aid the construction of effective interval training programs (i.e., intensity and duration of the work bouts and recovery intervals) for the enhancement of fitness, functional capacity, or cardiovascular and metabolic health (9,13,24). This study has also shown that the reconstitution of W' during the recovery intervals between repeated bouts of intense exercise results in a reduced rate of change of both iEMG and $\dot{V}O_2$ as exercise proceeds. This indicates that the depletion

of the W' and the associated increase in $\dot{V}O_2$ with time are portents of muscle fatigue development and important determinants of performance during intermittent exercise.

This study received no external funding.

The authors have no conflicts of interest. Weerapong Chidnok was supported by a Ph.D. scholarship from the National Science and Technology Development Agency of the Royal Thai Government.

The results of the present study do not constitute endorsement by the American College of Sports Medicine.

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